

STAT 548: Papers for PhD Research Training Course

Overview. My research concerns high-performance algorithms for Bayesian modeling and Probabilistic Machine Learning. An outcome of this research is the development of algorithms which support probabilistic programming languages such as STAN. These algorithms must be efficient, easy-to-use (“black box”), and general-purpose in order to accommodate a large range of models and to support a model development workflow. Two important classes of algorithms in this space are sampling methods, such as Markov chain Monte Carlo, and approximation methods, such as variational inference.

This research is at the intersection of theory, computation, and application. The papers I have selected demonstrate how to work at this intersection. When considering the topic of each paper, you might ask yourself:

- Would a practitioner using Bayesian modeling benefit from this paper’s contribution?
- Can the proposed method be implemented in a general-purpose statistical software package?
- Does the analysis in the paper allow us to answer these questions?

Paper list

- “Nested $\hat{R}$: Assessing the Convergence of Markov chain Monte Carlo when Running Many Short Chains.” Margossian et al. *Bayesian Analysis* (2024)
- “Parallelizing MCMC Across the Sequence Length.” Zoltowski et al. *NeurIPS* (2025)
- “Skew-symmetric Approximations of Posterior Distributions.” Pozza et al. *JRSS-B* (2026)
- “To select or not to select: predictively consistent priors instead of model selection.” Riha et al. *arXiv:2606.22850* (2026)

Evaluation.

- For each paper you will write a critical analysis which summarizes the contributions of the paper. Your analysis should (i) contextualize the paper and discuss recent literature, (ii) highlight the strengths and limitations as a reviewer might do, and (iii) discuss open questions and propose future research directions—ones which might actually lead to your own research project! If the paper has theoretical contributions, you should be able to reproduce the

proofs step-by-step and fill in gaps which may have been glossed over in the paper. If the paper proposes a new algorithm, you should be able to explain the associated pseudo-code and run the code associated with the paper. Finally, you should be able to explain in great detail the figures in the paper.

- In addition, we will come up with a small project for each paper. For example, you might extend one of the numerical experiments which evaluates the performance of a method across several models by adding a new model. In doing so, you will write code to implement the method, evaluate the performance and produce figures describing your findings. Another example: you might extend a theoretical result by coming up with a new conjecture, and either prove it, provide empirical evidence that supports the conjecture, or produce a counter-example.

As you work through the paper, I will be particularly interested in your ability to identify which parts of the paper you understand thoroughly and which ones remain unclear to you. I might grill you on some of the details in the papers with the goal of helping you develop the ability to grill yourself when you read papers. It's ok if some of the paper is unclear—these papers are challenging—and it is my expectation that we will spend time together clarifying various concepts.